DeformHG: Teacher-Guided Deformable Hypergraph Convolutional Neural Network for Semisupervised Hyperspectral Image Classification

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***Abstract*—Hyperspectral image classification (HSIC) remains challenging due to the complex high-order interactions among land-cover categories and the scarcity of labeled samples. Existing graph- and hypergraph-based methods usually rely on either fixed topologies or repeated topology reconstruction, which may induce structural instability and additional computational burden. To address these issues, we propose a deformable hypergraph convolutional neural network (Deform HGCN) and based on it, we put forward a semisupervised learning framework with a teacher–student architecture, termed DeformHG. The Deform HGCN adaptively learns hypergraph topology through feature-driven vertex–hyperedge association offsets and similarity-guided modulation. By incrementally updating the hypergraph without full reconstruction, Deform HGCN preserves topological continuity while maintaining computational efficiency. To further exploit unlabeled data, a pseudolabel refinement and mixed augmentation (PLRMA) module is developed to enhance pseudolabel reliability and data diversity under limited supervision. Experiments on four public hyperspectral datasets show that DeformHG delivers robust and competitive classification performance under limited labeled samples. The implementation will be publicly available at: https://github.com/liuwenp/DeformHG**

***Index Terms*—Deformable mechanisms, hypergraph convolutional networks, hyperspectral image classification (HSIC), semisupervised learning.**

I. Introduction

HYPERSPECTRAL image classification (HSIC) is an important task in remote sensing image interpretation [1] and has been widely applied to precise material identification [2], composition analysis [3], and medical diagnosis [4], among other domains [5]. Compared with RGB images, hyperspectral images (HSIs) contain rich and continuous spectral information, providing fine-grained discriminative cues for distinguishing different land-cover categories [6]. However, HSIC remains challenging because different categories may exhibit high spectral similarity, large intraclass variability, and complex spatial distributions. These factors lead to intricate spectral–spatial relationships among hyperspectral pixels. Meanwhile, pixel-level annotation usually requires expert knowledge and is costly to obtain, resulting in very limited labeled samples. Therefore, effectively modeling complex spectral–spatial relationships under label-scarce conditions is a critical problem in HSIC.

Early HSIC methods were based on probabilistic models and classical machine learning, such as spectral feature matching [7], support vector machines [8], and random forests [9], which struggle to capture nonlinear spectral–spatial patterns [10] and require handcrafted features. With the rapid advancement of image processing technology, deep learning-based HSIC methods have emerged as the mainstream approach [11]. These methods automatically extract hierarchical spectral–spatial features and achieve superior accuracy. Recent CNN and Transformer-based architectures have further improved spectral–spatial representation learning in HSIC. For example, SSSGAN [12] introduces a spectral–spatial synergy-guided attention mechanism to jointly exploit local spectral and spatial information, while MCMTN [13] explores multi-range spatial dependencies through convolutional modulation and Transformer architectures. However, their dependence on extensive labeled supervision remains a major limitation in label-scarce scenarios.

To address this limitation, semisupervised learning [14] has attracted considerable attention, as it exploits abundant unlabeled samples together with limited labeled data. Existing semisupervised methods commonly fall into four primary paradigms: low-density separation methods [15], graph-based methods [16], generative model-based methods [17], and contrastive learning [18]/self-training-based [19] methods.

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Among all paradigms, graph-based methods have become one of the most effective approaches due to their ability to capture the intrinsic manifold structure of HSI [20]. Early graph-based semisupervised HSIC can be traced back to the Gaussian-kernel graph framework proposed by Camps-Valls et al. [21], where data are treated as vertices, pairwise similarities define edges, and information is propagated over the graph. Graph neural networks (GNNs) inherit this structure and use learnable neural networks to update the graph representation [22]. Qin et al. [23] first introduced GNNs into semisupervised HSIC. Subsequently, miniGCN optimized graph construction and neighborhood aggregation strategies to alleviate the computational burden on large-scale HSI, establishing GNNs as a strong baseline [24].

However, conventional graph structures mainly describe pairwise relationships between vertices, making them insufficient for capturing high-order interactions among multiple spectrally and spatially related pixels [25]. To address this limitation, hypergraph neural networks (HGNNs) [26] have been introduced into HSIC [27], where hyperedges connect multiple related vertices and thus enable more expressive high-order spectral–spatial modeling. However, most existing HGNN-based methods rely on a preconstructed incidence matrix whose topology remains fixed during feature propagation, and thus may fail to reflect the evolving feature distribution. Dynamic hypergraph convolutional neural networks (DHGCNs) [28] improve topology adaptivity by reconstructing the hypergraph with updated vertex features, but full reconstruction may introduce abrupt topology variations and additional computational cost. This raises an important question: how can high-order vertex–hyperedge associations be adaptively updated in a continuous and stable manner without repeatedly reconstructing the entire hypergraph structure?

In parallel, deformable mechanisms, originating from deformable convolutional networks (DCNs) [29], offer a powerful way to handle irregular structures and geometric variations. By integrating learnable 2-D spatial offsets directly into the convolutional kernels, DCN dynamically adjusts the receptive field according to input content. Since HSIs have sophisticated land cover combinations, recent HSIC studies have explored deformable learning to improve spatial adaptability. For example, Xia et al. [30] extended deformable mechanisms to attention modules, Zhu et al. [31] first applied DCN in HSIC, and Zhong et al. [32] explored deformable Fourier filtering for hyperspectral feature learning. Nevertheless, existing deformable mechanisms are primarily defined on Euclidean image grids [33], where spatial adaptivity is achieved by adjusting sampling locations or receptive fields [34]. On the contrary, hypergraph convolution operates on non-Euclidean and discrete vertex–hyperedge associations, where "deformation" should not be interpreted as spatial coordinate shifts, but rather as continuous modulation of the incidence matrix.

Motivated by this, we propose a deformable hypergraph convolutional neural network (Deform HGCN) through a feature-driven hypergraph deformation module. This module adaptively learns a deformable hypergraph topology through vertex–hyperedge association adjustment. In particular, learnable association offsets are first introduced based on vertex features to obtain a preliminary deformed incidence matrix. This matrix is then refined by a similarity-guided modulation that controls the magnitude of each vertex–hyperedge adjustment. Significantly, the deformation is applied incrementally, updating only the previously constructed incidence matrix rather than reconstructing the entire hypergraph at each step. This incremental update strategy ensures that topology changes are bounded and smooth across layers, preserving both global structure and local adaptivity, and maintaining training stability.

Based on Deform HGCN and to tackle the limited labeled samples in HSIC, we further propose a semisupervised deformable hypergraph learning framework, DeformHG. DeformHG is built upon a teacher–student architecture, in which the teacher–student network jointly leverages Transformer and Deform HGCN to fully learn high-order spectral–spatial relations. In addition, a pseudolabel refinement and mixed augmentation (PLRMA) module is introduced to improve the reliability of unlabeled samples and enhance robustness under semisupervised training. By integrating deformable high-order relation modeling and reliable semisupervised learning into a unified framework, DeformHG can learn robust spectral–spatial representations under limited labeled data.

This work makes the following contributions.

1) We propose a Deform HGCN that formulates deformable hypergraph learning as continuous vertex–hyperedge association adjustment through feature-driven association offsets and similarity-guided modulation.
2) We develop a PLRMA module to mitigate noise accumulation during the pseudolabeling generation stage, thereby providing more reliable and diverse supervision in semisupervised learning.
3) We formulate a unified semisupervised optimization framework and conduct extensive experiments on multiple HSI datasets. The results confirm the effectiveness of DeformHG in terms of classification accuracy and training stability.

The remainder of this article is structured as follows. Section II presents DeformHG, Section III introduces the experiment settings and results, Section IV analyzes the effects of different parameters, and Section V provides a conclusion.

II. Method

A. Overall Framework

DeformHG implements semisupervised HSIC through a teacher–student architecture, which could extract knowledge from both labeled and unlabeled data, as depicted in Fig. 1. First, we obtain a coordinate-augmented HSI $\mathbf{X} \in \mathbb{R}^{H\times W\times(B+2)}$ by concatenating spatial indices with the original HSI, where H, W, and B represent the height, width, and band number, respectively. The spatial indices are derived from the pixelwise spatial coordinates of the HSI. Then, spectral feature extraction is performed on the augmented HSI, yielding labeled spectral data and unlabeled spectral data $\mathbf{X}^{\mathrm{l}}_{\mathrm{spe}}, \mathbf{X}^{\mathrm{u}}_{\mathrm{spe}} \in \mathbb{R}^{1\times 1\times(B+2)}$,

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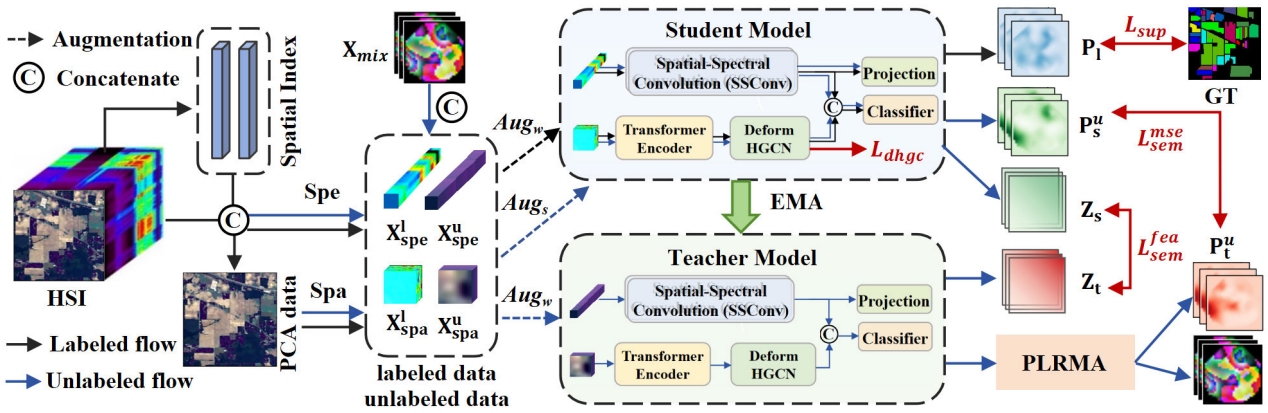


Fig. 1. Overall framework of DeformHG. Aug_w and Aug_s, respectively, denote the weak augmentation and the strong augmentation. P^u_t and Z_t indicate the teacher model's probability prediction and corresponding latent feature representation, respectively. P^u_s and Z_s indicate the student model's probability prediction and corresponding latent feature representation, respectively.

which are then utilized in the following semisupervised learning framework. Meanwhile, principal component analysis (PCA) is applied to the coordinate-augmented HSI for dimensionality reduction and reducing spectral redundancy. The reduced data are represented as $\mathbf{X} \in \mathbb{R}^{H\times W\times D}$, where D denotes the number of principal components retained after PCA. Subsequently, spatial feature extraction is conducted on the reduced data to obtain labeled spatial patches and unlabeled spatial patches $\mathbf{X}^{\mathrm{l}}_{\mathrm{spa}}, \mathbf{X}^{\mathrm{u}}_{\mathrm{spa}} \in \mathbb{R}^{p\times p\times D}$, where p denotes the patch size, which is set to 16 in our implementation.

During training, labeled data undergo only a weak augmentation transformation before being fed into the student model, while unlabeled data undergo both weak augmentation and strong augmentation simultaneously. Based on these predictions, the supervised loss L_{sup} is computed between predictions and ground-truth labels, the hypergraph regularization loss L_{dhgc} is computed on the supervised data, and consistency regularization L^{mse}_{sem} is enforced between teacher and student outputs, along with a contrastive loss L^{fea}_{sem} to enhance feature discrimination. Notably, after the teacher model, a PLRMA module is employed to mitigate error propagation from noisy pseudolabels and enhance training stability. During inference, no data augmentation is applied. The test data are directly fed into the network to produce the final classification results.

B. Network Architecture

We adopt the mean teacher framework [35] for semisupervised optimization. In this framework, the teacher and student models are initialized with the same network structure, while their update mechanisms are different. The student model is trained with supervision from labeled samples and refined pseudolabels, whereas the teacher model is updated by the exponential moving average (EMA) of the student parameters. This update strategy effectively smooths temporal evolution and provides stable supervision.

To fully exploit the spectral–spatial characteristics of HSI, the network is designed with two parallel branches to extract spectral and spatial features, respectively. In particular, spectral features F_{spe} are extracted using two SSConv modules, where SSConv is composed of a 1×1 convolution and a 2-D grouped convolution. The extraction of spatial feature F_{spa} employs a Transformer encoder and a Deform HGCN, as shown in Fig. 2. Transformer captures long-range spatial contextual dependencies, while the Deform HGCN models adaptive high-order vertex–hyperedge relationships.

C. Transformer Encoder

Considering the limited availability of labeled data in HSIC, we introduce a lightweight Transformer variant by stacking two encoder layers. The input spatial data $\mathbf{X}_{\mathrm{spa}} \in \mathbb{R}^{p\times p\times D}$ is first projected into embedding sequences via a patch embedding layer. Subsequently, multihead self-attention is introduced to model global contextual relationships. Each attention head focuses on different types of feature interactions, enabling richer and more comprehensive feature modeling. The attention operation can be mathematically expressed as follows:

$$\mathrm{Attention}\,(\mathbf{Q},\mathbf{K},\mathbf{V}) = \mathrm{Softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \tag{1}$$

where the $\mathbf{Q}$, $\mathbf{K}$, and $\mathbf{V}$ matrices are obtained by linearly projecting the input features $\mathbf{X}$ through three learnable transformation matrices, denoted as $\mathbf{W}_{\mathrm{q}}$, $\mathbf{W}_{\mathrm{k}}$, and $\mathbf{W}_{\mathrm{v}}$, respectively. d_k represents the dimensionality of $\mathbf{K}$. Repeat the above attention operation multiple times and concatenate the resulting outputs to form the final multihead attention representation

$$\mathrm{Multihead}\,(\mathbf{X}) = \mathrm{Concat}\,(\mathrm{head}_1,\ldots,\mathrm{head}_h)\,\mathbf{W}^o \tag{2}$$

$$\mathrm{head}_i = \mathrm{Attention}\ (\mathbf{Q},\mathbf{K},\mathbf{V})\,. \tag{3}$$

Here, $\mathbf{W}^o$ denotes a learnable parameter matrix.

D. Deformable Hypergraph Convolutional Neural Network

1) Hypergraph Convolutional Neural Network: The hypergraph structure is constructed based on the relationships among vertices. For each vertex n, its k-nearest neighbors are selected according to feature similarity to form a hyperedge p. This construction allows hyperedges to encode high-order associations beyond pairwise connections. The hypergraph is represented using an incidence matrix $\mathbf{H} \in \mathbb{R}^{|N|\times|P|}$, where $|N|$

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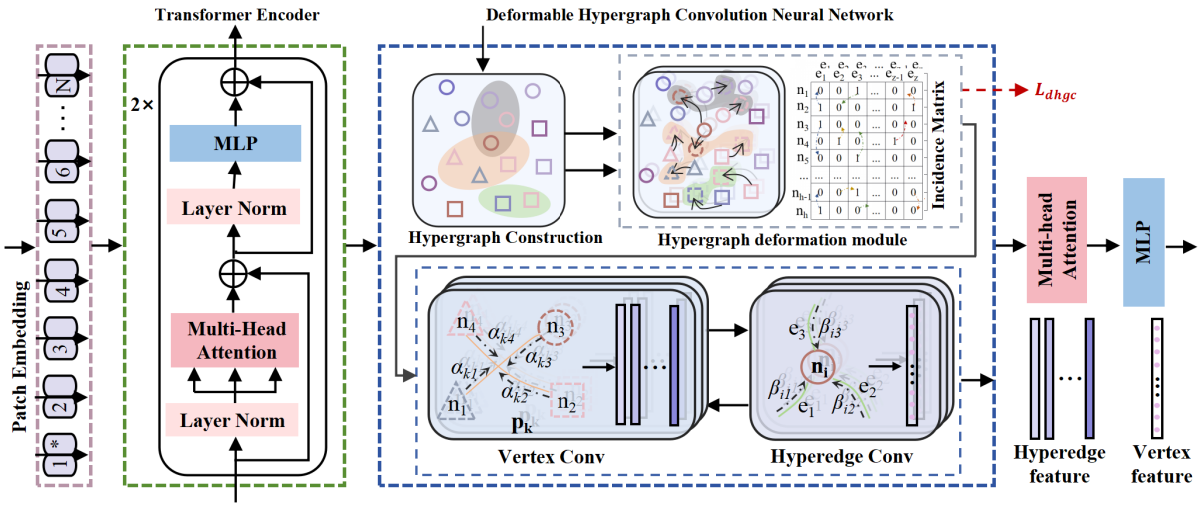


Fig. 2. Architecture of the spatial feature F_{spa} extraction network.

denotes the number of vertices and $|P|$ denotes the number of hyperedges. For a given vertex n and a hyperedge p, H can be expressed as

$$\mathbf{H}_{n,p} = \begin{cases} 1, & n \in p \\ 0, & n \notin p. \end{cases} \tag{4}$$

Once the hyperedge weight matrix W and the incidence matrix $\mathbf{H}$ are defined, we can derive the vertex degree matrix D_n and hyperedge degree matrix D_p. Based on the constructed hypergraph, feature extraction is performed via hypergraph convolution. In particular, given vertex features $\mathbf{N}$ as input, the hypergraph convolution can be formulated as

$$\hat{\mathbf{N}} = \sigma\left(D_n^{-1/2}\mathbf{H}WD_p^{-1}\mathbf{H}^{\top}D_n^{-1/2}\mathbf{N}\mathbf{\Theta}^{l}\right) \tag{5}$$

where $\hat{\mathbf{N}}$ denotes the updated vertex features, $W \in \mathbb{R}^{|P|\times|P|}$ represents the hyperedge weight matrix, which is set as the identity matrix for simplicity. $\mathbf{\Theta}^{l}$ is a learnable parameter and σ is a nonlinear activation function.

Hypergraph convolution consists of two steps: the vertex convolution transformation mechanism (Vertex Conv) and the hyperedge convolution transformation mechanism (Hyperedge Conv), as illustrated in Fig. 2. First, Vertex Conv aggregates vertex features to form hyperedge features. In particular, the input vertex features are first transformed via a learnable multilayer perceptron (MLP) to obtain a transformation matrix. This transformation matrix is then multiplied by the input vertex features to update the hidden representation of each hyperedge. Then, Hyperedge Conv propagates hyperedge features back to the vertices to update vertex representations. Concretely, an MLP assigns a weight to each hyperedge connected to a vertex n, and the weighted hyperedge features are then aggregated to update the vertex features.

2) Deformable Hypergraph Module: Different from DHGCNs that reconstruct the entire hypergraph according to the updated vertex representations, i.e., $\mathbf{H}^{l}$ = Reconstruct($\mathbf{N}^{l}$), where $\mathbf{N}^{l}$ denotes the vertex feature representation at the lth layer. Deform HGCN retains the previously constructed hypergraph information and adaptively deforms its vertex–hyperedge associations according to the current vertex representations. As shown in Fig. 3, the hypergraph deformation module performs a two-stage deformation process. In the first stage, a feature-driven association offset is introduced in an incremental manner to update the existing hypergraph associations. In the second stage, similarity-guided modulation is applied according to feature-level compatibility to further refine the deformation magnitude.

The first stage adapts the topology by modifying the incidence values while retaining the existing vertex and hyperedge sets. In addition, we first enhance the previously constructed hypergraph representation using a multihead attention mechanism. The learned association offset is then added to the attention-enhanced hypergraph representation, which is further transformed by an MLP followed by layer normalization to obtain the preliminary deformed incidence matrix

$$\mathbf{H}_p^{l} = \mathrm{LN}\left[\mathrm{MLP}\left(\mathbf{H}^{l-1} + \mathrm{MHA}\left(\mathbf{H}^{l-1}\right) + \Delta\mathbf{H}^{l}\right)\right] \tag{6}$$

where $\mathbf{H}_p^{l}$ denotes the preliminary deformed incidence matrix, $\Delta\mathbf{H}^{l}$ represents the learnable vertex–hyperedge association offset. Since the sparse incidence matrix $\mathbf{H}$ primarily encodes discrete vertex–hyperedge associations and does not directly reflect the evolving feature distribution, we predict the association offset from the current vertex representations

$$\Delta\mathbf{H}^{l} = \phi\left(\mathbf{BN}\left(C_{1\times1}\left(\mathbf{AvgPool}\left(\mathbf{N}^{l}\right)\right)\right)\right) \tag{7}$$

where **AvgPool** denotes average pooling, $C_{1\times1}$ denotes a 1×1 convolution, BN$(\cdot)$ represents batch normalized, and $\phi(\cdot)$ represents the GELU nonlinear activation function.

To adaptively control the deformation strength, the cosine similarity and the Euclidean-distance-based similarity are computed between vertex features and the preliminary deformed vertex–hyperedge representations. The two similarity measures are fused to generate a similarity-guided modulation matrix $\mathbf{M}^{l}$, which regulates the deformation magnitude of each vertex–hyperedge association

$$\mathbf{M}^{l} = \sigma\left(f_m\left([S_{cos}, S_{euc}]\right)\right) \tag{8}$$

$$\widetilde{\mathbf{H}}^{l} = \mathrm{Norm}\left(\mathbf{M}^{l} \odot \mathbf{H}_p^{l}\right) \tag{9}$$

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where $f_m(\cdot)$ is implemented as a learnable linear transformation that takes the concatenated similarity vector $[S_{cos}, S_{euc}]$ as input, σ represents the sigmoid activation function, $\odot$ indicates elementwise multiplication, and Norm$(\cdot)$ is the normalization operation. Modulation $\mathbf{M}^l$ is applied to the incidence matrix before hypergraph convolution, thereby changing high-order message-passing paths rather than merely reweighting output features. This design enhances reliable vertex–hyperedge associations and suppresses noisy or redundant associations, yielding a more adaptive incidence matrix for subsequent hypergraph convolution.

It is worth noting that introducing multiple deformable processes may increase the degrees of freedom of topology learning, which can lead to unstable training if not properly constrained. In particular, excessive deformation may cause vertex–hyperedge associations to become overly concentrated, while different hyperedges may produce redundant or highly overlapping association patterns. To address this issue, we impose a hypergraph regularization loss on the deformed HGCN

$$\begin{aligned}\mathcal{L}_{dhgc} &= \mathcal{L}_{sparsity} + \mathcal{L}_{diversity} \\ &= \|\widetilde{H}\|_1 + \sum_{i\neq j} \left|\widetilde{p}_i^{\top}\widetilde{p}_j\right| \end{aligned} \tag{10}$$

where $\mathcal{L}_{sparsity}$ is an L_1-based sparsity penalty that encourages sparse vertex–hyperedge associations and prevents overconcentrated topology deformation. $\mathcal{L}_{diversity}$ is a hyperedge diversity penalty that promotes the uniqueness of different hyperedges and reduces redundant association patterns; $\widetilde{p}_i^{\top}$ is the vectorized representation corresponding to the ith hyperedge of $\widetilde{\mathbf{H}}$. By jointly enforcing sparsity and diversity, $\mathcal{L}_{dhgc}$ stabilizes the optimization of the deformable hypergraph module and improves the reliability of topology adaptation.

E. PLRMA Module

In semisupervised learning, unreliable pseudolabels may accumulate during training and mislead subsequent optimization. To address this problem, we develop a PLRMA module, which improves the reliability of unlabeled samples through three complementary mechanisms: historical consistency, conflict-aware pseudolabel correction, and mixed augmentation. The detailed procedure is summarized in Algorithm 1.

First, historical consistency is employed to maintain a prediction history R_u and C_u for each unlabeled sample. Only predictions with confidence c higher than a predefined threshold T_c are stored, which filters out low-confidence noise and provides reliable historical supervision. Based on the historical records, a confidence-weighted voting strategy is used to obtain a stable historical pseudolabel m, reducing the oscillation caused by single-step predictions. Second, conflict-aware pseudolabel correction is introduced to handle inconsistency between the current prediction and the historical pseudolabel. When a conflict occurs, PLRMA first verifies the reliability of the historical pseudolabel through the consistency threshold T_m, and then compares the current confidence c with the reliability of the historical prediction and selects the more trustworthy label as the final pseudolabel $\hat{r}$. In this way, unreliable current predictions can be corrected by historical information, while highly confident new predictions are still allowed to update outdated pseudolabels. The highly confident new predictions are defined by the current c exceeding αT_c. Here, $\alpha > 1$ imposes a stricter confidence requirement to mitigate pseudolabel oscillation. α is empirically set to 1.1 in all experiments. Third, after obtaining the final label $\hat{r}$, a MixUp-style strategy is adopted to generate mixed data X_{mix} between labeled data X_l and reliable unlabeled data. This operation increases data diversity, smooths decision boundaries, and provides more diverse supervision for semisupervised training.

Algorithm 1 PLRMA

Require: Unlabeled data X_u; current prediction label r and confidence c; History sets R_u, C_u; confidence threshold T_c; consistency threshold T_m; Labeled data X_l; Current epoch e; Total epochs E

Ensure: Final label $\hat{r}$; Mixed data X_{mix}

$T_c \leftarrow T_c^0 + \frac{e}{E} \times 0.2$ // Confidence threshold

$T_m \leftarrow T_m^0 + \frac{e}{E} \times 0.2$ // Consistency threshold

// Step 1: High-confidence filtering

if $c > T_c$ **then**

$R_u \leftarrow R_u \cup \{r\}, C_u \leftarrow C_u \cup \{c\}$

// Step 2: Compute reliable historical pseudo-label

for all class k **do**

$m \leftarrow \arg\max_k (C_u^{\top} \cdot \mathbf{1}(R_u = k))$

$n_m \leftarrow \frac{\mathbf{1}(R_u=m)^{\top}\mathbf{1}}{|R_u|}$ // Historical consistency ratio

if $n_m < T_m$ **then**

$m \leftarrow$ None

// Step 3: Conflict-aware selection

$$\hat{r} \leftarrow \begin{cases} r, & \text{if } r = m \text{ or } c \geq \alpha T_c \\ m, & \text{otherwise} \end{cases}$$

// Step 4: Mix augment

if Reliable pseudo-label $\hat{r}$ exists **then**

Randomly pick $x_l \in X_l$ with label $\hat{r}$, and pick $x_u \in X_u$

Sample $\lambda \sim$ Uniform$(0.8, 1.0)$

$X_{mix} \leftarrow \lambda \cdot x_l + (1-\lambda) \cdot x_u$

Furthermore, different from dash [36], PLRMA adopts a gradually increasing dynamic threshold strategy; both T_c and T_m are adaptively adjusted during training. In the early stages, when predictions are unstable, lower thresholds are adopted to accelerate the accumulation of historical information while leveraging the auditing mechanism to filter erroneous pseudolabels. As training progresses and predictions become more reliable, the thresholds are gradually increased to reduce unnecessary auditing overhead and avoid overcorrection, thereby leading to more stable training dynamics.

F. Overall Loss

We employ multiple loss functions for the model optimization, including supervised loss, hypergraph regularization loss, consistency regularization loss, and feature contrastive loss. The supervision loss is formulated with cross-entropy loss and

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contains two parts: L_{sup}^{l} measures the classification error on labeled data with ground-truth annotations and L_{sup}^{mix} supervises the mixed virtual data using their corresponding pseudolabels. The total supervised loss of the model is

$$L_{sup} = L_{sup}^{l} + L_{sup}^{mix}. \tag{11}$$

This combination loss is particularly beneficial to the model in label-scarce scenarios.

The consistency regularization loss calculates the mean squared error (mse) L_{sem}^{mse} between predictions from strongly and weakly augmented inputs, encouraging the probability distribution produced by the student model to directly approximate that of the teacher model. In particular, to reduce computational cost while minimizing the propagation of erroneous information, we first input the teacher model's predictions into the PLRMA module when computing L_{sem}^{mse}, effectively filtering out noisy data. Subsequently, L_{sem}^{mse} is used to align the probability distributions predicted by the teacher and student models. The loss can be formulated as

$$L_{sem}^{mse} = \frac{1}{K}\sum_{k=1}^{K}\left(\mathrm{P}_{\mathrm{t_i},\mathrm{k}}^{u} - \mathrm{P}_{\mathrm{s_i},\mathrm{k}}^{u}\right)^2 \tag{12}$$

Here, $\mathrm{P}_{\mathrm{t_i},\mathrm{k}}^{u}$ denotes the probability distribution predicted by the teacher model for the ith sample and the kth class and $\mathrm{P}_{\mathrm{s_i},\mathrm{k}}^{u}$ denotes the probability distribution predicted by the student model for the same sample.

Furthermore, to improve training robustness, we introduce a feature contrastive loss L_{sem}^{fea}. In particular, we measure the cosine similarity between the predictions extracted from the teacher and student models to construct a positive–negative sample matching matrix M

$$M_{ij}^{\mathrm{pos}} = \mathbb{I}\left(S_{ij} \geq \tau_{\mathrm{pos}}\right), \quad M_{ij}^{\mathrm{neg}} = \mathbb{I}\left(S_{ij} \leq \tau_{\mathrm{neg}}\right) \tag{13}$$

where S_{ij} represents the cosine similarity between the teacher-derived and student-derived predictions. M_{ij}^{pos} is positive sample matching matrix, M_{ij}^{neg} is negative sample matching matrix, τ_{pos} is the positive threshold, and τ_{neg} is the negative threshold. Following the strategy adopted in contrastive mutual learning with seudolabel smoothing (CMLPL) [37], we set τ_{pos} to 0.8 and τ_{neg} to 0.3 for all datasets.

Since semantic similarity across different augmented views should correspond to similarity in the feature space, this constraint encourages semantically similar data to be closer in the feature space while pushing dissimilar data farther apart. Therefore, we leverage feature contrastive loss to extract this invariant feature representation. We regard pseudolabel-consistent data as the positive pairs and view the pseudolabel-inconsistent data as the negative pairs. The contrastive loss can be expressed as follows:

$$L_{\mathrm{feat}}^{s\to t} = -\frac{1}{N}\sum_{i=1}^{N}\log\left(\frac{\sum_{j=1}^{N} M_{ij}^{\mathrm{pos}}\cdot\exp\left(\mathrm{Q}_{ij}\right)}{\sum_{j=1}^{N}\left(M_{ij}^{\mathrm{pos}} + M_{ij}^{\mathrm{neg}}\right)\cdot\exp\left(\mathrm{Q}_{ij}\right)}\right). \tag{14}$$

Among them, N is the total number of data within a batch. The feature similarity matrix $\mathrm{Q}_{ij} = \mathrm{Z}_{\mathrm{s_i}}\cdot\mathrm{Z}_{\mathrm{t_j}}^{\mathrm{T}}$. The feature similarity constraint is $L_{sem}^{fea} = 1/2(L_{\mathrm{feat}}^{s\to t} + L_{\mathrm{feat}}^{t\to s})$.

We also introduce an additional hypergraph regularization loss L_{dhgc} on the supervised data, which has been described in detail in Section II-D2 deformable hypergraph module. The total loss of our model is

$$L_{\mathrm{total}} = L_{sup} + \gamma_{mse}\cdot L_{sem}^{mse} + \gamma_{fea}\cdot L_{sem}^{fea} + L_{dhgc}. \tag{15}$$

III. Experiments

A. Datasets

To assess the performance of DeformHG, we evaluate it on four widely used public datasets. The University of Pavia dataset was acquired by the ROSIS sensor with a spatial resolution of 1.3 m and contains 103 spectral bands. It represents a typical urban scene and has a spatial size of 610×340 pixels and includes nine land-cover classes. The Liao-Ning 02 dataset was a fusion dataset of hyperspectral and multispectral imagery, both acquired by the ZY-1 02-D satellite. The fusion dataset is a wetland scene with a spatial resolution of 2.5 m and 144 spectral bands. The dataset size is 1536×1536 pixels, including nine land-cover classes. The Indian Pines dataset was collected by the AVIRIS sensor with a spatial resolution of 20 m and consists of 200 spectral bands. It represents a mixed agricultural scene and has a size of 145×145 pixels, containing 16 land-cover classes. The Houston 2013 dataset was captured by the ITRES CASI-1500 sensor with a spatial resolution of 2.5 m and contains 144 spectral bands. It represents an urban scene and has a size of 349×1905 pixels, including 15 land-cover classes. The detailed land-cover categories and dataset splits are listed in Table I.

B. Experiment Settings

The training process is performed for 25 epochs, where the patch size and batch size are set to 16 and 128, respectively. Adam is adopted as the optimizer, and the teacher model is updated from the student model using an EMA with a momentum coefficient of 0.95. The initial learning rate is set to 0.0005, with a weight decay of 0.0001. For each dataset, 15 labeled samples per class are used for training. If the class contains fewer than 30 samples, half of the available samples in that class are used for training. In addition, we adopt a transductive semisupervised learning setting, where 10 000 unlabeled samples are randomly selected from the entire image, whose ground-truth labels are not accessed during training. The per-class accuracy (PA), overall accuracy (OA), average accuracy (AA), and kappa coefficient (K) are used to conduct a comprehensive evaluation of different models' performance. To further compare computational efficiency, we report the training time, inference time, and model parameters of each method. All experiments are implemented with Python 3.8.20 and PyTorch 1.10.0, and conducted on a workstation equipped with an Intel i7-10700K CPU, 32 GB of RAM, and an NVIDIA GeForce RTX 3090 GPU.

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TABLE I

LAND COVER CATEGORIES AND DATASET DIVISION FOR THE UNIVERSITY OF PAVIA, LIAO-NING 02, INDIAN PINES, AND HOUSTON 2013 DATASETS

University of Pavia dataset					Liao-Ning 02 dataset			Indian Pines dataset			Houston 2013 dataset		
Class	Color	Land Cover	Training	Testing	Land Cover	Training	Testing	Land Cover	Training	Testing	Land Cover	Training	Testing
1		Asphalt	15	6616	Liao river	15	12731	Alfalfa	15	31	Healthy grass	15	1236
2		Meadows	15	18634	Phragmites austrails	15	56593	Corn-notill	15	1413	Stressed grass	15	1239
3		Gravel	15	2084	Paddy fields	15	37013	Corn-mintill	15	815	Synthetic grass	15	682
4		Trees	15	3049	Suaeda salsa	15	26329	Corn	15	222	Trees	15	1229
5		Metal Sheets	15	1330	Intertidal muds	15	70318	Grass-pasture	15	468	Soil	15	1227
6		Bare Soil	15	5014	Reed pond	15	7530	Grass-trees	15	715	Water	15	310
7		Bitumen	15	1315	Intertidal muds	15	4058	Grass-pasture-mowed	14	14	Residential	15	1253
8		Bricks	15	3667	Aquaculture ponds	15	6945	Hay-windrowed	15	463	Commercia	15	1229
9		Shadows	15	932	Industrial land	15	1117	Oats	10	10	Road	15	1227
10		-	-	-	-	-	-	Soybean-notill	15	957	Highway	15	1212
11		-	-	-	-	-	-	Soybean-mintill	15	2440	Railway	15	1220
12		-	-	-	-	-	-	Soybean-clean	15	578	Parking Lot 1	15	1218
13		-	-	-	-	-	-	Wheats	15	190	Parking Lot 2	15	454
14		-	-	-	-	-	-	Woods	15	1250	Tennis Court	15	413
15		-	-	-	-	-	-	Bldg-Grass-Trees-Drivers	15	371	Running Track	15	630
16		-	-	-	-	-	-	Stone-Stell-Towers	15	78	-	-	-
		Total	**135**	**42641**	**Total**	**135**	**222634**	**Total**	**234**	**10015**	**Total**	**225**	**14804**

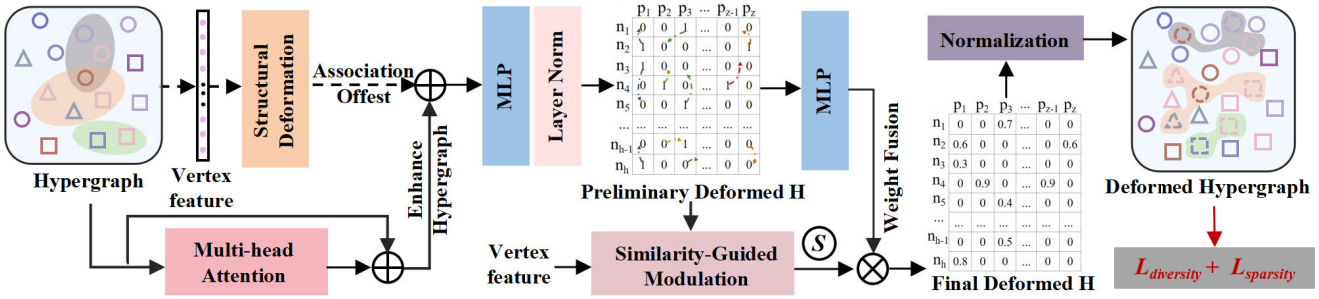


Fig. 3. Structure of the deformable hypergraph module.

TABLE II

QUANTITATIVE RESULTS OF VARIOUS MODELS ON THE UNIVERSITY OF PAVIA DATASET

Class	CNN	ViT-DGCN [38]	ViT-DHGC	Semi-CNN [39]	DMSGer [40]	RSEN [41]	CMLPL [37]	DSCA [42]	RMAE [43]	SSMM [44]	Semi-HCN [45]	DeformHG
1	77.21±11.11	73.95±5.66	77.88±5.82	84.92±4.14	76.90±2.30	84.92±4.14	77.75±0.16	84.12±2.75	73.04±0.02	97.30±0.22	99.12±0.56	**99.65±0.18**
2	98.21±0.48	74.64±10.91	91.36±2.80	97.83±2.96	91.78±2.79	91.19±2.96	97.33±0.50	98.66±0.81	97.49±0.91	84.04±1.09	90.16±0.60	**99.66±0.28**
3	58.40±7.72	69.65±1.72	89.68±5.39	70.56±1.81	86.21±4.80	**96.88±1.81**	95.65±0.40	93.55±1.70	81.24±0.05	96.29±0.97	84.58±14.67	96.65±0.55
4	58.08±10.76	85.76±1.20	85.48±2.62	49.82±9.95	90.79±2.46	92.40±1.95	91.65±0.17	93.96±0.30	91.67±1.73	98.83±0.09	97.12±0.61	**99.33±0.07**
5	85.68±6.01	99.14±0.79	99.94±0.09	95.74±0.19	99.95±0.09	99.76±0.62	99.99±0.01	99.70±0.88	97.74±0.16	**100.00±0.00**	**100.00±0.00**	99.47±0.07
6	90.36±2.49	93.78±0.71	97.77±1.41	98.35±0.86	84.73±4.35	97.91±1.25	93.88±0.99	**98.76±0.04**	96.35±0.48	97.00±0.49	97.02±2.09	88.15±1.83
7	58.90±2.68	97.39±0.15	98.80±1.57	81.01±0.00	98.81±0.64	**100.00±0.00**	**100.00±0.00**	99.94±0.15	99.39±0.07	98.07±0.24	96.50±2.87	99.19±0.60
8	58.73±12.13	90.56±3.74	68.37±11.49	88.17±4.98	91.34±3.37	68.96±4.98	68.45±2.09	**99.81±0.39**	90.48±0.34	94.63±2.10	96.56±1.88	94.98±1.45
9	38.40±8.51	97.33±2.24	90.34±6.36	94.86±1.20	88.61±5.30	98.93±1.19	99.17±0.09	92.62±1.57	76.07±5.30	99.83±0.05	90.39±5.84	**99.98±0.04**
OA (%)	80.24±3.98	83.47±4.02	85.42±2.23	86.98±2.13	88.66±3.60	90.16±1.40	91.12±0.28	94.91±4.08	91.23±0.26	91.45±0.49	93.64±0.64	**97.72±0.69**
AA (%)	72.91±3.44	80.82±0.82	78.70±0.96	84.75±0.63	89.90±4.89	92.33±0.78	91.54±0.20	95.68±0.14	88.95±1.23	96.22±0.16	94.61±1.57	**97.43±0.73**
K×100	74.52±4.75	79.23±4.68	84.41±2.81	82.93±2.62	88.50±3.40	87.21±1.75	88.31±0.35	90.79±0.23	88.43±0.35	88.97±0.61	91.69±0.85	**96.96±0.93**
Train (s)	**13.01**	18.63	23.62	130.10	2204.69	168.27	302.90	6845.96	5824.21+44.66	125.96	109.32	403.17
Test (s)	203.86	105.25	133.21	1.26	**0.45**	4.25	28.89	3.01	3.98	1.98	1.74	6.45
Para (k)	64.79	244.09	247.09	90.58	89.94	1236.49	552.33	1816.34	207.11	**60.73**	67.09	285.05

To comprehensively evaluate the effectiveness of DeformHG, we compare it with 11 representative methods, including both supervised CNN and semisupervised CNN-based (Semi-CNN) [39] methods, as well as the supervised ViT-DGCN [38] and a ViT-DHGC model. To explicitly assess the effectiveness of the Deform HGCN, the ViT-DHGC model is derived from ViT-DGCN by replacing the dynamic GCN module with Deform HGCN while keeping all other components unchanged. In addition, we include semisupervised dynamic graph convolution network (DMSGer) [40], self-ensembling network (RSEN) [41], CMLPL [37], dual-stream class-adaptive network (DSCA) [42], regularized masked autoencoder (RMAE) [43], semisupervised manifold method (SSMM) [44], and hierarchical classification network (Semi-HCN) [45]. All experiments are conducted using the hyperparameters recommended in the respective original papers to ensure fair comparison. Each experiment is independently repeated ten times, and the mean value and standard deviation are reported to ensure reliable evaluation.

C. Classification Results

Tables II–V present the quantitative classification results for the four datasets, while Figs. 4–7 illustrate the qualitative classification maps. Overall, DeformHG achieves the best

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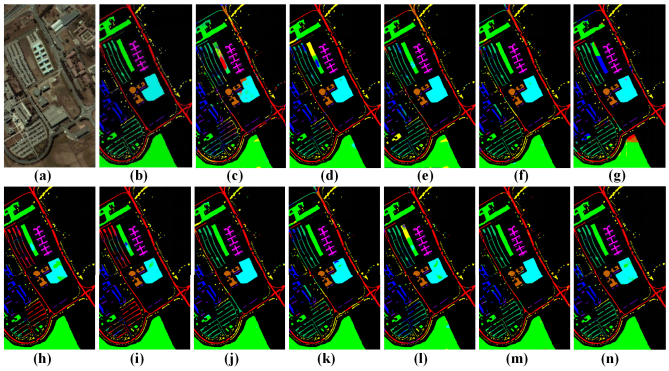


Fig. 4. Qualitative results of various models on the University of Pavia dataset. (a) False-color image. (b) Ground truth map. (c) CNN. (d) ViT-DGCN. (e) ViT-DHGC. (f) Semi-CNN. (g) DMSGer. (h) RSEN. (i) CMLPL. (j) DSCA. (k) RMAE. (l) SSMM. (m) Semi-HCN. (n) DeformHG.

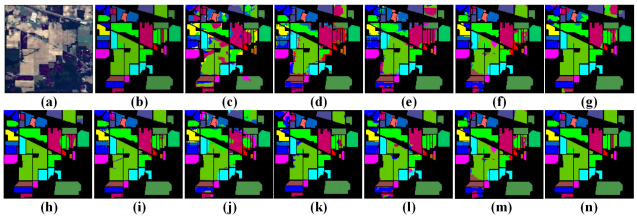


Fig. 5. Qualitative results of various models on the Indian Pines dataset. (a) False-color image. (b) Ground truth map. (c) CNN. (d) ViT-DGCN. (e) ViT-DHGC. (f) Semi-CNN. (g) DMSGer. (h) RSEN. (i) CMLPL. (j) DSCA. (k) RMAE. (l) SSMM. (m) Semi-HCN. (n) DeformHG.

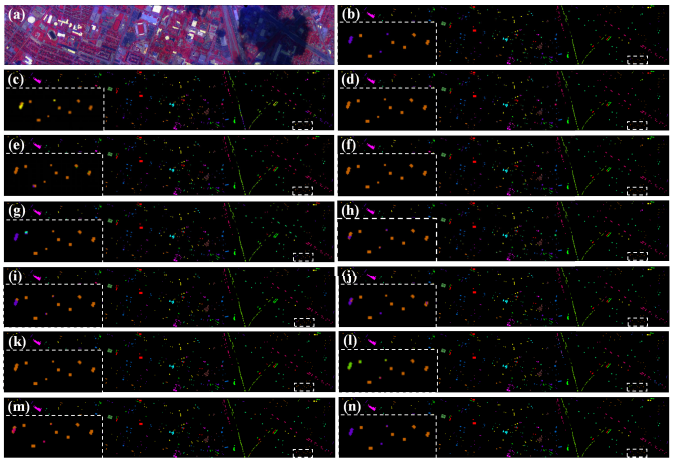


Fig. 6. Qualitative results of various models on the Houston 2013 dataset. (a) False-color image. (b) Ground truth map. (c) CNN. (d) ViT-DGCN. (e) ViT-DHGC. (f) Semi-CNN. (g) DMSGer. (h) RSEN. (i) CMLPL. (j) DSCA. (k) RMAE. (l) SSMM. (m) Semi-HCN. (n) DeformHG.

classification accuracy across all datasets and demonstrates good stability in terms of standard deviation in the quantitative results. In terms of computational complexity, DeformHG exhibits moderate complexity compared with the competing approaches. For the qualitative results, the classification maps generated by DeformHG show the closest agreement with the ground-truth land-cover distribution.

The University of Pavia dataset, characterized by its high spatial resolution and abundant fine-grained land cover details, has been widely established as a canonical benchmark for the

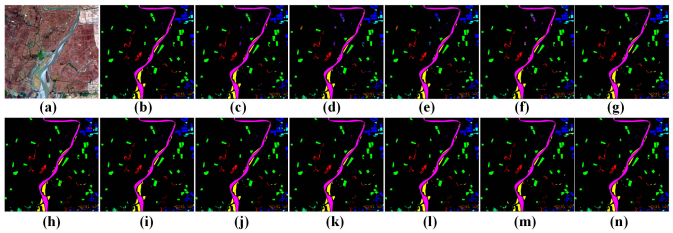


Fig. 7. Qualitative results of various models on the Liao-Ning 02 dataset. (a) False-color image. (b) Ground truth map. (c) CNN. (d) ViT-DGCN. (e) ViT-DHGC. (f) Semi-CNN. (g) DMSGer. (h) RSEN. (i) CMLPL. (j) DSCA. (k) RMAE. (l) SSMM. (m) Semi-HCN. (n) DeformHG.

quantitative evaluation of HSIC algorithms. As presented in Table II, although DeformHG exhibits slight misclassifications in Class 6, its OA reaches 97.72%, yielding a 2.81% improvement over the suboptimal method DSCA. Qualitatively, as illustrated in Fig. 4, compared to the distinct salt-and-pepper noise generated by other methods in continuous homogeneous regions and object edges, the classification map produced by DeformHG demonstrates high internal coherence and the most precise boundary delineation. Furthermore, in-depth analysis reveals that for highly similar and easily confused challenging samples, such as asphalt, the PA of DeformHG is significantly improved, demonstrating the model's robust feature discrimination capability. This performance leap is attributed to the core architectural design: compared with the ViT-DHGC model and ViT-DGCN, the proposed Deform HGCN can more exhaustively mine topological discriminative information from limited samples than standard dynamic graphs. Meanwhile, the introduction of the semisupervised paradigm effectively captures the underlying structure of unlabeled data, successfully mitigating model oscillation and performance degradation caused by pseudolabel noise accumulation. Finally, regarding computational efficiency, DeformHG requires fewer than 0.3M parameters and compresses the training time to 7 min, achieving an excellent balance between lightweight design and high performance. Notably, the total training time of RMAE is the sum of the training durations of its encoder and classifier, as the two modules are trained separately.

The Indian Pines dataset is one of the earliest and most widely recognized canonical benchmark datasets. As reported in Table III, most methods generally yield lower classification accuracy on this dataset due to its low spatial resolution and severe mixed-pixel phenomena. However, the OA of DeformHG reaches 99.51%, achieving a clear margin over all compared methods. This effectively demonstrates the powerful feature extraction capability of DeformHG, demonstrating remarkable stability and robustness in label-scarce semisupervised scenarios. In the qualitative results, DeformHG outputs the classification map that most closely resembles the ground truth. Regarding PA, DeformHG achieves over 97% for every class, ensuring balanced classification across categories. Significantly, for challenging classes such as corn-notill, soybean-mintill, and soybean-clean, which are notoriously difficult for other methods to identify, DeformHG achieves a significant performance breakthrough. In terms of lightweight design, while DeformHG may not have the absolute

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TABLE III
QUANTITATIVE RESULTS OF VARIOUS MODELS ON THE INDIAN PINES DATASET

Class	CNN	ViT-DGCN [38]	ViT-DHGC	Semi-CNN [39]	DMSGer [40]	RSEN [41]	CMLPL [37]	DSCA [42]	RMAE [43]	SSMM [44]	Semi-HCN [45]	DeformHG
1	**100.00±0.00**	**100.00±0.00**	99.35±1.29	85.16±7.24	**100.00±0.00**	96.15±1.89	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	98.75±2.50	66.90±36.78	97.42±0.61
2	48.76±2.41	51.93±3.87	56.93±6.71	81.12±3.79	66.95±2.50	83.66±2.20	88.05±2.25	82.45±2.64	67.87±2.11	84.54±4.42	76.67±19.88	**99.53±0.11**
3	62.93±7.94	68.34±5.30	72.91±4.84	62.65±2.41	76.78±4.82	92.22±1.59	89.11±2.12	93.01±2.01	65.52±2.05	85.33±2.98	87.97±1.89	**99.29±0.15**
4	92.07±3.57	95.50±2.07	94.77±6.03	61.26±0.95	98.38±1.94	76.96±2.14	98.16±2.21	98.20±0.43	99.10±0.00	97.29±1.32	**99.82±0.36**	99.10±0.56
5	75.90±1.42	81.15±1.64	82.91±1.52	94.49±0.65	93.38±0.54	97.62±0.84	94.13±5.27	97.86±1.59	81.84±1.22	96.51±1.38	89.10±3.67	**98.80±0.16**
6	94.11±2.31	92.73±2.81	93.68±3.45	97.65±0.15	83.19±1.19	97.75±0.28	99.67±0.22	97.90±0.29	90.49±1.08	99.11±0.64	96.63±3.11	**99.97±0.00**
7	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	68.00±0.69	78.18±29.09	**100.00±0.00**
8	98.97±2.05	99.83±0.21	99.74±0.25	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.35±0.89	**100.00±0.00**
9	**100.00±0.00**	96.00±8.16	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
10	79.31±1.88	81.76±2.83	78.75±4.76	70.76±2.17	85.08±1.82	80.67±1.73	87.55±4.24	93.63±1.74	73.15±2.85	84.42±3.37	93.40±2.52	**98.87±0.14**
11	66.56±1.25	71.08±2.88	71.65±2.47	76.10±2.41	77.50±1.25	76.14±8.71	89.64±3.56	88.40±0.64	69.75±2.39	86.81±5.47	90.32±3.73	**99.39±0.06**
12	59.79±4.92	62.46±5.60	82.98±3.56	65.12±3.83	86.99±2.39	65.09±2.36	83.57±6.58	90.31±1.18	89.27±2.38	89.68±2.83	89.72±3.84	**99.58±0.26**
13	99.59±0.60	99.58±0.84	99.58±0.52	99.58±0.21	97.58±1.65	99.46±0.69	99.49±0.30	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.47±0.82	**100.00±0.00**
14	95.59±1.11	98.06±0.96	95.73±1.18	82.74±3.02	86.16±0.97	90.36±2.04	99.29±0.89	71.12±0.27	96.16±1.14	90.98±2.03	97.18±3.09	**100.00±0.00**
15	98.94±1.04	98.44±2.46	99.30±0.50	73.91±3.39	99.68±0.31	98.91±0.27	99.75±0.29	94.88±3.44	93.53±0.75	99.44±0.59	94.47±9.36	**100.00±0.00**
16	94.46±3.86	99.49±0.63	98.72±0.81	98.97±0.96	99.23±0.63	**100.00±0.00**	99.37±1.05	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	98.68±0.83	**100.00±0.00**
OA (%)	74.76±1.20	77.85±1.01	79.83±2.32	79.20±0.59	82.44±5.36	85.30±2.41	92.26±0.83	88.88±4.03	79.71±1.12	89.36±1.19	90.63±2.57	**99.51±0.08**
AA (%)	85.41±0.76	82.14±0.64	89.19±1.49	84.34±0.84	90.68±4.25	90.85±0.73	95.49±0.44	94.23±3.31	88.71±12.78	94.07±0.49	86.95±5.23	**99.50±0.12**
K×100	71.97±1.38	75.06±1.09	77.32±2.58	76.27±0.66	80.22±5.67	83.29±2.72	91.15±0.94	87.39±3.98	77.11±1.24	87.87±1.35	89.29±2.99	**99.44±0.10**
Train (s)	**15.40**	32.95	41.98	109.01	499.62	170.15	299.99	19216.94	1334.15+22.71	90.09	65.69	363.00
Test (s)	22.99	17.46	19.40	0.38	**0.08**	0.67	3.36	1.03	1.08	0.83	0.31	1.08
Para (k)	**65.16**	244.54	247.55	105.73	235.52	1339.41	608.14	1821.73	207.57	74.63	82.88	299.91

TABLE IV
QUANTITATIVE RESULTS OF VARIOUS MODELS ON THE HOUSTON 2013 DATASET

Class	CNN	ViT-DGCN [38]	ViT-DHGC	Semi-CNN [39]	DMSGer [40]	RSEN [41]	CMLPL [37]	DSCA [42]	RMAE [43]	SSMM [44]	Semi-HCN [45]	DeformHG
1	81.98±0.44	92.30±3.76	**99.42±0.41**	85.88±2.38	86.12±3.05	95.94±1.38	90.45±3.58	96.04±0.29	90.53±3.20	93.05±1.32	95.22±1.84	97.39±0.38
2	86.99±10.13	93.64±3.18	96.32±3.10	96.62±1.05	83.88±5.34	98.71±0.79	93.70±6.32	87.97±0.34	93.38±0.38	95.50±3.75	96.62±1.88	**99.35±0.32**
3	97.76±2.10	98.30±0.62	98.62±0.81	99.12±0.17	99.38±0.41	98.59±2.38	**100.00±0.00**	98.68±0.14	98.39±0.27	98.78±1.95	95.35±7.65	**100.00±0.00**
4	84.83±1.05	70.53±1.32	72.50±1.20	94.32±0.80	83.24±5.54	97.87±0.11	99.35±5.44	89.34±0.88	67.78±6.64	98.75±0.33	93.84±0.50	**99.98±0.03**
5	99.11±0.32	99.97±0.40	99.77±0.13	96.83±0.00	99.38±0.57	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.92±0.34	**100.00±0.00**	95.44±2.00	**100.00±0.00**
6	87.03±4.22	87.48±0.52	88.52±2.73	97.12±0.40	**100.00±0.00**	96.32±1.88	98.06±1.80	82.26±9.89	86.77±0.33	88.25±0.49	94.42±2.35	99.42±0.13
7	84.44±5.11	84.45±2.38	77.57±5.31	69.90±6.01	87.81±4.65	87.65±0.81	85.40±6.60	87.55±0.42	84.12±5.82	91.89±3.09	73.54±4.75	**98.52±0.85**
8	50.85±5.57	72.42±2.19	74.48±1.05	62.82±5.65	59.90±6.72	72.53±1.53	75.59±1.07	73.07±1.01	89.02±0.27	66.26±1.51	73.89±9.41	**97.48±0.97**
9	56.70±24.86	61.62±3.78	68.26±5.12	82.90±3.68	67.86±4.12	88.63±0.38	84.24±8.87	86.01±0.24	71.71±7.05	86.51±2.92	91.82±4.34	**99.18±0.37**
10	69.93±2.41	96.49±3.51	99.34±0.43	83.46±2.57	85.73±9.84	92.26±4.55	**100.00±0.00**	99.34±0.32	97.19±0.00	92.08±2.32	93.52±7.07	**100.00±0.00**
11	65.16±11.10	97.33±2.07	97.34±1.09	84.79±2.30	94.65±4.53	88.70±1.14	91.06±7.80	94.92±0.68	97.13±0.02	73.22±0.50	91.71±2.30	**99.89±0.10**
12	53.22±5.41	74.17±1.41	81.41±5.32	47.17±11.85	84.53±5.73	78.13±1.71	85.79±5.86	72.33±0.90	89.33±0.97	69.78±7.39	49.74±3.56	**99.26±0.34**
13	88.47±7.65	84.67±4.88	78.59±14.05	93.65±2.65	98.15±0.58	74.85±10.36	75.11±4.46	**99.78±0.64**	91.85±1.36	44.27±6.27	96.11±4.78	96.08±1.11
14	98.82±0.78	99.32±0.32	99.56±0.54	**100.00±0.00**	**100.00±0.00**	99.47±0.28	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.76±0.21	99.22±1.10	**100.00±0.00**
15	**100.00±0.00**	97.36±0.83	99.69±0.33	**100.00±0.00**	99.94±0.12	**100.00±0.00**	**100.00±0.00**	97.98±0.56	**100.00±0.00**	**100.00±0.00**	99.94±0.12	**100.00±0.00**
OA (%)	77.15±1.79	85.98±0.49	87.88±1.05	83.50±1.15	86.03±4.92	90.81±0.35	91.33±0.73	90.04±3.10	89.36±0.25	86.96±1.08	87.51±1.15	**99.12±0.09**
AA (%)	80.35±1.90	81.88±0.55	83.21±1.28	86.31±0.91	88.70±3.22	91.31±0.66	92.73±0.59	91.02±2.11	90.28±9.07	86.54±1.00	89.36±1.00	**99.10±0.08**
K×100	75.31±1.95	84.85±0.53	86.90±1.13	82.21±1.24	84.91±4.74	90.06±0.37	90.63±0.79	89.24±2.11	88.49±0.27	85.90±1.17	86.52±1.25	**99.05±0.09**
Train (s)	**12.91**	23.07	36.77	132.93	1527.20	140.51	254.95	13452.00	2267.03+85.75	238.62	286.28	413.27
Test (s)	24.72	15.48	15.55	10.66	**0.46**	12.59	48.61	1.51	1.60	3.25	3.97	24.79
Para (k)	**65.13**	244.48	247.48	94.76	176.54	1281.55	608.14	1844.57	207.50	72.54	75.15	292.30

minimum parameter count, it optimally achieves a comprehensive balance of "high accuracy, lightweight architecture, and fast convergence."

On the Houston 2013 dataset, as shown in Table IV, DeformHG achieves an OA of 99.12%, and the PA for each class remains above 96%; furthermore, DeformHG simultaneously achieves high OA, AA, and kappa values with low standard deviations, indicating that DeformHG provides highly stable and balanced classification in label-scarce scenarios. Fig. 6 also visually confirms the precision of DeformHG's classification boundaries. Regarding computational complexity, although DeformHG has a slight gap compared to extremely lightweight methods, its absolute accuracy advantage of over 7% grants it significant overall competitiveness.

The Liao-Ning 02 dataset is a fused dataset located in an urban–rural ecological ecotone. As shown in Table V, this dataset is relatively less challenging, and most methods achieve satisfactory classification results. Under few-shot conditions, DeformHG still attains a high classification performance of 99.72%. In Fig. 7, DeformHG demonstrates superior capability in distinguishing spectrally similar targets such as paddy fields, reed ponds, reeds, and seepweeds, suggesting the adaptability of Deform HGCN to the morphological variations of complex land covers.

IV. DISCUSSION AND ANALYSIS

A. Ablation Study

To thoroughly validate the effectiveness of each core component in DeformHG, as demonstrated in Table VI, we progressively introduce the teacher–student semisupervised learning scheme and the PLRMA module into the supervised baseline. Here, TS denotes the teacher–student semisupervised learning scheme, PLR denotes pseudolabel refinement, and mix denotes mixed augmentation. All configurations are trained with the supervised loss on labeled samples. The experimental setting remains identical in all ablation experiments. The semisupervised learning scheme significantly outperforms

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TABLE V
QUANTITATIVE RESULTS OF VARIOUS MODELS ON THE LIAO-NING 02 DATASET

Class	CNN	ViT-DGCN [38]	ViT-DHGC	Semi-CNN [39]	DMSGer [40]	RSEN [41]	CMLPL [37]	DSCA [42]	RMAE [43]	SSMM [44]	Semi-HCN [45]	DeformHG
1	76.85±2.20	86.19±1.49	86.97±1.23	89.90±2.89	95.91±1.95	96.39±2.90	**99.21±0.73**	98.77±0.69	93.35±3.12	98.86±0.14	92.97±0.09	98.55±0.16
2	87.91±6.13	89.42±4.88	89.98±4.55	90.56±1.55	89.70±2.21	99.44±0.56	99.97±0.16	**99.99±0.01**	98.97±0.28	99.45±0.19	99.57±0.44	98.81±0.07
3	80.72±6.68	94.00±3.34	93.58±3.03	88.26±2.41	96.85±1.54	97.23±5.36	**99.97±0.41**	99.20±0.21	99.62±0.14	99.79±0.13	99.64±0.02	99.93±0.07
4	94.27±1.76	94.94±0.61	94.87±1.04	96.70±0.58	98.46±0.40	98.46±1.27	97.68±0.11	98.85±0.55	96.77±1.21	99.49±0.43	99.12±0.82	**99.78±0.13**
5	97.54±0.40	97.46±2.16	97.66±2.01	99.85±0.11	99.71±0.17	99.27±0.49	99.93±0.27	98.83±0.31	99.93±0.25	99.91±0.01	**99.97±0.06**	99.94±0.04
6	93.90±4.12	96.62±3.62	96.88±3.63	94.78±0.74	84.71±11.66	90.71±0.66	93.85±0.74	99.55±0.30	**99.83±0.06**	68.06±10.56	93.11±5.66	99.32±0.22
7	95.27±3.74	96.68±1.86	96.67±2.15	90.01±1.22	98.28±1.53	98.99±0.06	98.30±0.22	99.37±0.26	98.64±0.22	93.91±2.42	98.67±2.85	**99.51±0.15**
8	99.20±0.47	99.16±0.55	99.09±0.58	99.68±0.21	**99.77±0.01**	99.45±0.55	99.77±0.22	99.59±0.11	99.44±0.14	90.15±0.86	97.05±6.53	99.68±0.09
9	92.46±3.75	93.59±6.53	93.52±6.15	95.63±0.85	97.27±0.40	99.06±0.04	**100.00±0.00**	99.33±0.34	97.94±0.50	99.19±0.27	88.33±0.15	99.93±0.15
OA (%)	93.24±2.10	93.89±0.89	94.07±1.14	94.24±0.64	95.78±0.52	98.43±0.72	99.40±0.41	99.24±0.05	98.83±0.34	98.17±0.37	98.95±0.18	**99.72±0.03**
AA (%)	83.91±1.51	84.81±0.75	84.92±0.85	87.26±0.43	95.63±1.30	97.90±0.43	**99.24±0.54**	**99.24±0.04**	98.28±0.17	94.31±1.42	96.49±0.79	98.63±0.02
K×100	91.50±2.59	92.33±1.09	92.55±1.41	92.69±0.65	94.68±0.65	98.01±0.90	98.01±0.52	98.99±0.10	98.52±0.45	97.67±0.47	98.62±0.42	**99.64±0.04**
Train (s)	**23.13**	31.78	42.73	187.05	5645.50	160.67	389.20	51907.70	29038.72+12.62	204.61	262.56	497.27
Test (s)	3218.31	2715.42	2582.46	45.69	**0.72**	50.51	105.49	10.28	23.88	5.29	5.31	80.90
Para (k)	**64.79**	244.09	247.09	95.96	172.92	1306.49	588.62	1839.95	207.11	70.55	72.42	290.38

TABLE VI
QUANTITATIVE RESULTS OF THE ABLATION OF EACH MODULE ON FOUR DATASETS

TS	PLR	Mix	University of Pavia			Indian Pine			Houston 2013			Liao-Ning 02		
			OA(%)	AA(%)	K×100	OA(%)	AA(%)	K×100	OA(%)	AA(%)	K×100	OA(%)	AA(%)	K×100
-	-	-	89.66±0.66	93.17±0.49	89.32±1.00	86.70±1.90	90.74±0.81	85.15±2.91	91.63±0.62	92.94±0.48	90.16±1.15	96.24±1.24	95.97±1.43	96.11±1.14
✓	-	-	92.93±1.06	94.39±1.10	90.69±1.35	92.96±2.36	96.38±0.83	91.96±2.68	95.14±0.37	95.52±0.36	94.74±0.40	97.52±0.93	97.12±1.07	97.29±0.89
✓	✓	-	96.83±0.50	96.64±0.31	95.78±0.64	99.39±0.10	99.40±0.15	99.25±0.08	98.73±0.15	98.80±0.15	98.63±0.16	99.24±0.25	98.74±0.85	99.05±0.38
✓	-	✓	95.87±0.39	95.88±0.33	94.50±0.48	98.05±0.37	98.81±0.24	97.63±0.42	96.32±0.59	96.70±0.55	95.86±0.63	99.03±0.53	98.72±0.69	99.00±0.43
✓	✓	✓	**97.72±0.69**	**97.43±0.73**	**96.96±0.93**	**99.51±0.08**	**99.50±0.12**	**99.44±0.10**	**99.12±0.09**	**99.10±0.08**	**99.05±0.09**	**99.72±0.03**	**99.24±0.04**	**99.64±0.04**

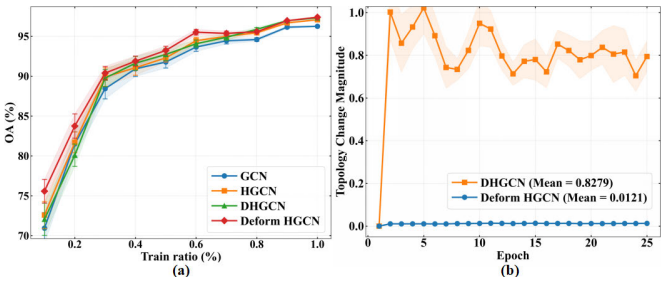


Fig. 8. (a) Average OA of different graph models on the University of Pavia dataset. (b) Average topology change magnitudes of Deform HGCN and DHGCN with different epochs on the University of Pavia dataset.

TABLE VII
COMPUTATIONAL COMPLEXITY OF DIFFERENT GRAPH MODELS

Method	Params(K)	FLOPs(M)	Memory(MB)	Train(s)	Infer(s)
GCN	168	2.17	38	55.80	1.13
HGCN	172	2.26	39	56.10	1.85
Deform HGCN	260	5.98	237	115.46	2.47
DHGCN	298	19.72	336	133.38	2.78

the fully supervised counterpart. This is attributed to the fact that semisupervised learning can exploit the intrinsic feature information contained in massive unlabeled data, which alleviates the severe overfitting problem caused by scarce labeled data in HSIC. With the introduction of PLR, the classification performance of DeformHG is improved. This improvement stems from the enhanced reliability of pseudolabels, which effectively reduces the propagation of erroneous information during model training, thus enabling the model to learn more discriminative and robust spectral–spatial features from HSI. The mix augmentation strategy further mixes unlabeled data with labeled data, which enriches the diversity of training samples and expands the effective training sample size. This strategy boosts the model's classification performance and generalization ability. The complete framework achieves the highest classification accuracy among all ablation variants, which demonstrates the effectiveness and rationality of each designed module in DeformHG.

B. Compare Different Graph Models

To evaluate the effectiveness of the proposed Deform HGCN, we design four classifiers based on GCNs, HGCNs, DHGCNs, and Deform HGCN, respectively. These experiments isolate the contribution of the deformable hypergraph mechanism by excluding the teacher–student framework and PLRMA module. Experiments are conducted on the University of Pavia dataset using different training ratios. The results are presented in Fig. 8(a). To further quantify the computational cost introduced by deformable topology learning, we conducted computational complexity analyses on different graph methods under the same experimental setting, and the results are shown in Table VII. As observed, Deform HGCN achieves competitive performance, with particularly significant improvements in low-sample regimes. This advantage can be attributed to its capability to integrate high-order relational modeling with adaptive vertex–hyperedge association learning, thereby overcoming the limitations of fixed relational structures. In particular, Deform HGCN enables the adaptive learning of both association offsets and the modulation matrix, allowing it to better capture irregular land-cover shapes and distributions, which are difficult to model using fixed graph structures. Although Deform HGCN introduces additional computational costs compared with HGCN, these overheads mainly come from association offset prediction and similarity-guided modulation. Different from DHGCN, which

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TABLE VIII
QUANTITATIVE RESULTS OF DIFFERENT LOSSES ON FOUR DATASETS

L_{sup}^{l}	L_{sem}^{mse}	L_{sem}^{fea}	L_{sup}^{mix}	University of Pavia			Indian Pine			Houston 2013			Liao-Ning 02		
				OA(%)	AA(%)	K×100	OA(%)	AA(%)	K×100	OA(%)	AA(%)	K×100	OA(%)	AA(%)	K×100
✓	-	-	-	91.86±0.88	93.32±0.42	90.92±1.02	88.22±1.69	92.29±0.75	86.60±1.90	93.12±0.47	93.74±0.50	92.57±0.58	96.92±0.89	95.34±1.05	96.64±0.62
✓	✓	-	-	95.84±0.29	95.94±0.23	95.13±0.36	99.18±0.11	99.39±0.03	99.06±0.09	98.53±0.21	98.60±0.15	98.43±0.24	99.17±0.73	98.78±0.85	98.94±0.94
✓	✓	✓	-	96.80±0.40	**96.70±0.31**	96.41±0.45	99.32±0.08	99.40±0.05	99.19±0.07	98.65±0.06	98.67±0.04	98.56±0.10	99.56±0.17	99.12±0.36	99.37±0.28
✓	✓	✓	✓	**97.17±0.36**	96.66±0.32	**96.22±0.49**	**99.51±0.08**	**99.50±0.12**	**99.44±0.10**	**99.12±0.09**	**99.10±0.08**	**99.05±0.09**	**99.72±0.03**	**99.24±0.04**	**99.64±0.04**

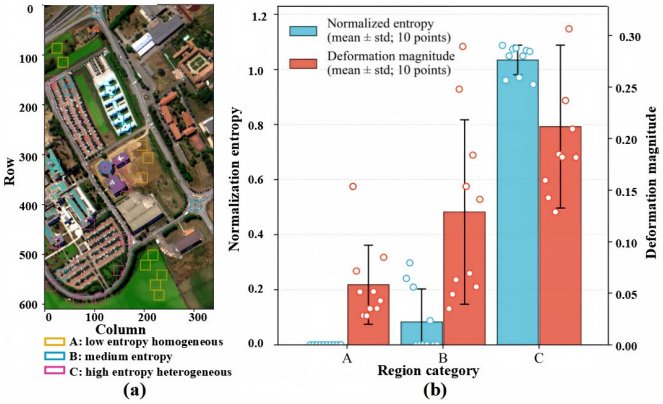


Fig. 9. (a) University of Pavia region selection. (b) Normalized entropy and deformation magnitude.

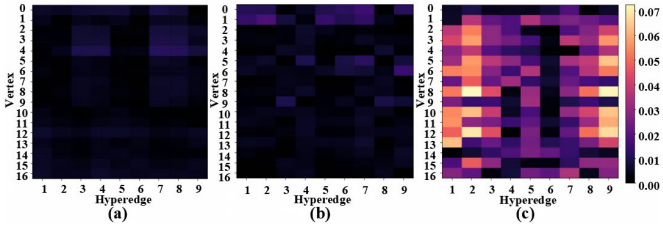


Fig. 10. Heatmap of deformation magnitude for the three example regions. (a) Region A. (b) Region B. (c) Region C.

reconstructs the entire topology at each iteration, Deform HGCN achieves joint adaptivity in association offsets and modulation, providing an adaptive topology learning strategy with reduced reconstruction overhead.

In addition, to analyze the continuity of topology updates, we visualize the magnitude of topology changes for DHGCN and Deform HGCN over ten independent runs. In particular, we computed the Frobenius norm of the difference between the $\mathbf{H}^{l}$ obtained at two consecutive epochs. As shown in Fig. 8(b), DHGCN exhibits pronounced peaks and fluctuations, indicating that each full reconstruction introduces substantial topology variation. On the contrary, the curve for Deform HGCN is nearly flat and significantly lower in magnitude, illustrating that its incremental updates produce smoother and more controlled topology changes.

C. Region Analysis of Hypergraph Deformation

To investigate whether Deform HGCN learns adaptive vertex–hyperedge association offsets rather than uniform feature reweighting, we analyze the deformation magnitudes in three land-cover regions with different spatial heterogeneity

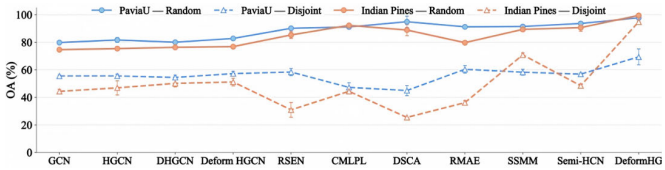


Fig. 11. Average OA of DeformHG and representative comparison methods under both random and spatially disjoint splits.

levels on the University of Pavia dataset. Fig. 9 shows the selected regions and the corresponding normalized Shannon entropy and deformation magnitude averaged over ten independent runs, while Fig. 10 presents the deformation magnitude heatmaps. All regions are fixed to 20×20 pixels for fair comparison.

The spatial heterogeneity of each region is quantified using the normalized Shannon entropy of the ground-truth class distribution. Region A represents a homogeneous vegetation-dominated region, Region B represents a relatively homogeneous artificial-surface region, and Region C represents a highly heterogeneous urban-boundary region.

As shown in Figs. 9 and 10, Region A exhibits the lowest entropy and deformation magnitude, while Region C shows the highest values. Region B presents an intermediate magnitude. These results indicate that the learned association offsets are not uniformly applied but adaptively vary with local structural complexity and feature heterogeneity. Stronger deformation magnitudes in heterogeneous regions suggest that Deform HGCN can better adjust vertex–hyperedge associations to model complex spatial–spectral relationships.

D. Spatial-Independent Evaluation

The original experiments follow a random pixel-level sampling protocol. Since neighboring pixels in HSI usually exhibit strong spatial correlations, we further evaluate DeformHG and representative comparison methods under a spatially disjoint protocol on the University of Pavia and Indian Pines datasets, following recent spatial-independent evaluation studies [46], [47].

As shown in Fig. 11, all methods suffer performance degradation under the spatially disjoint setting, indicating that spatial distribution shifts introduce a more challenging evaluation scenario. This also suggests that random pixel-level sampling may lead to optimistic performance estimation due to spatial dependencies between training and testing samples.

Nevertheless, DeformHG maintains relatively smaller performance degradation compared with other methods, demonstrating its robustness under stricter spatial-independent eval-

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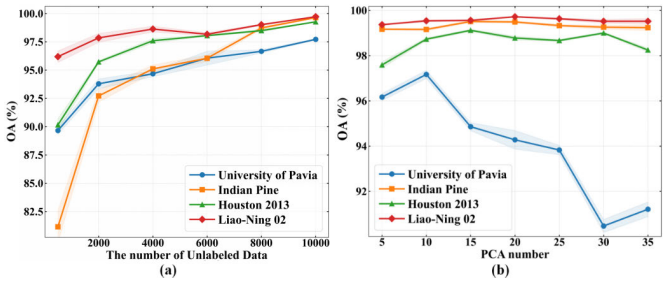


Fig. 12. Average OA of DeformHG on each dataset. (a) Different numbers of unlabeled data. (b) Different number of PCA.

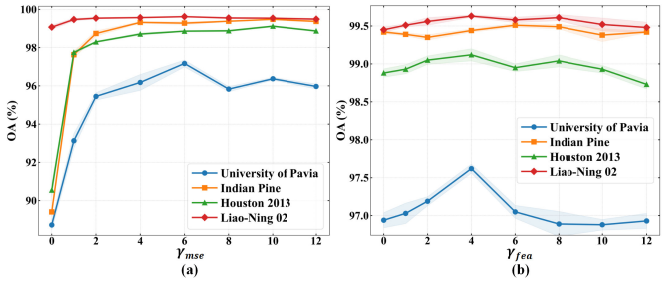


Fig. 13. Average OA of DeformHG with different loss function weights on each dataset. (a) Different γ_{mse}. (b) Different γ_{fea}.

uation. This advantage is attributed to its adaptive high-order spatial–spectral relationship modeling, which enables more effective structural representation under spatial distribution variations.

E. Parameter Analysis

1) Number of Unlabeled Data: Fig. 12(a) illustrates that classification accuracy improves as the number of unlabeled data increases and eventually converges to a stable level. This improvement occurs because unlabeled data provide valuable information about the intrinsic geometric structure and probability distribution in the feature space. Such information helps the model correct potential biases or incorrect assumptions that may arise when learning solely from sparse labeled data. In essence, labeled data inform the model what is correct, whereas unlabeled data reveal what is plausible in the underlying data distribution. But the improvement gradually slows down as the amount of unlabeled data increases and eventually converges to a stable level. This phenomenon occurs because the decision boundary approaches its near-optimal position, and additional unlabeled data can no longer provide new structural information to further refine it.

2) Number of PCA: In this study, we employ PCA for spectral dimensionality reduction to eliminate data redundancy. To achieve the balance between information preservation and model performance, we analyzed the impact of the number of dimensions of HSI on the classification performance. We incrementally increased the number of retained principal components from 5 to 35. As shown in Fig. 12(b), due to the relatively distinct spectral differences and high band redundancy among land cover classes in the University of Pavia dataset, the model peaks in performance at a dimensionality of 10; further increasing the dimensions introduces noise and triggers overfitting. On the contrary, the Indian Pines dataset suffers from severe mixed pixels, while the Houston 2013 and Liao-Ning 02 datasets exhibit high spectral similarity among land covers. Consequently, they achieve their optimal classification performance at retained dimensions of 15, 15, and 20, respectively.

3) Loss Function Weight: The losses of DeformHG differ substantially in terms of numerical scale, optimization targets, and convergence behavior. To evaluate the effectiveness of each component, we conduct an ablation study on the loss functions, as reported in Table VIII. To prevent any single loss from dominating the gradient updates and to ensure stable joint optimization of multiple objectives, we assign weights to each semisupervisory signal to regulate their contribution. In our experiments, the weights of supervised loss L_{sup} and hypergraph regularization loss L_{dhgc} are fixed at 1, while the weights of the semisupervised consistency loss γ_{mse} and feature contrastive loss γ_{fea} are varied for analysis.

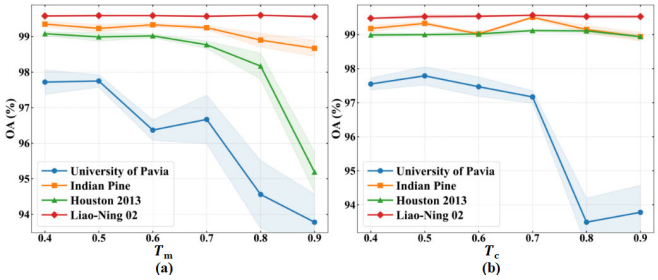


Fig. 14. Average OA of DeformHG with different thresholds on each dataset. (a) Different T_{m}. (b) Different T_{c}.

The corresponding results are presented in Fig. 13. As the weight of the consistency loss increases, the classification performance on the four datasets first improves and then gradually declines. This behavior is attributed to the fact that the semisupervised consistency loss is a key component of semisupervised learning and typically contributes positively to model training. However, when its value becomes excessively large, the model tends to overemphasize semantic alignment rather than making predictions, which may weaken the discriminative learning required for classification. On the contrary, variations in the feature contrastive loss have a relatively smaller impact on performance. The optimal values of γ_{mse} are 6, 10, 10, and 6 on the University of Pavia, Indian Pines, Houston 2013, and Liao-Ning 02 datasets, respectively. For the feature contrastive loss, γ_{fea} = 4 yields the best overall performance across all four datasets.

4) Threshold Value: In PLRMA, the settings of confidence threshold T_{c} and consistency threshold T_{m} are critical to the module's efficacy: if T_{c} and T_{m} are set too high, the module fails to accumulate sufficient effective history, rendering the review mechanism invalid. Conversely, excessively low thresholds introduce massive noise and frequently trigger reviews, impairing training efficiency. To determine the optimal baseline thresholds, we conducted a systematic parameter sensitivity analysis. As shown in Fig. 14, the experiments indicate that the best values (T_{c}, T_{m}) for the Pavia University dataset are (0.5, 0.4); Indian Pines and Houston 2013 peak at

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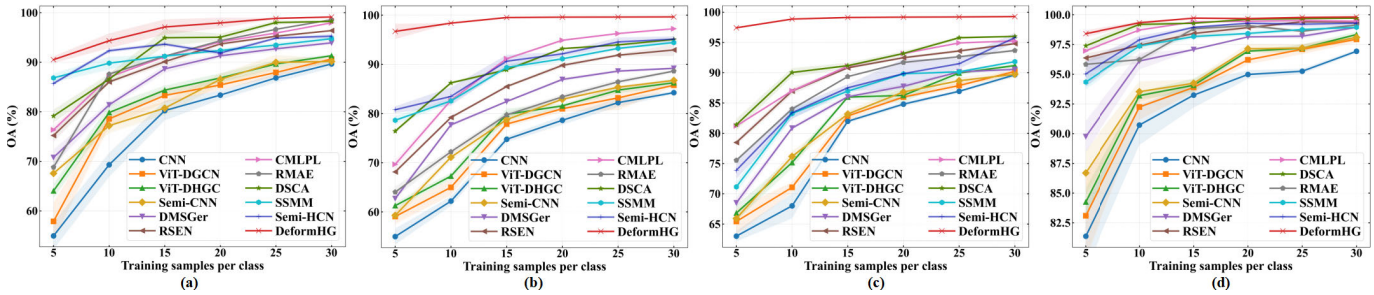


Fig. 15. OA of each model with different quantities of training labeled samples for each class. (a) University of Pavia. (b) Indian pine. (c) Houston 2013. (d) Liao-Ning 02.

(0.7, 0.4); while the optimal combination for the Liao-Ning dataset is (0.7, 0.6).

The variation in optimal thresholds across different datasets intrinsically stems from the heterogeneity of their physical characteristics and interclass separability. In particular, the University of Pavia dataset exhibits significant intraclass variance due to its extremely high spatial resolution. Therefore, a relatively lower T_c is adopted to inclusively absorb these challenging yet valid samples, ensuring the effective accumulation of historical priors. On the contrary, for the severe mixed-pixel phenomenon in Indian Pines and the highly confusing spectral similarity in Houston 2013, DeformHG must enforce a higher entry barrier to strictly filter out noise. Furthermore, for the Liao-Ning 02 dataset, which is relatively simple, DeformHG is expected to maintain highly stable predictions. Consequently, alongside a high T_c, the most stringent T_m is applied to rigorously eliminate any disruptive outliers that lack historical support.

F. Generalization Performance

To evaluate DeformHG's generalization performance under different training sample sizes, we conduct experiments with different numbers of labeled data for all compared methods. As illustrated in Fig. 15, DeformHG consistently maintains strong classification performance across different training sample settings. In the case of limited samples, the OA of DeformHG far exceeds that of other methods, which is attributed to the sufficient feature extraction of the semisupervised teacher–student architecture and the reasonable design of the network architecture. However, as the number of labeled data continues to increase, the performance gap between different methods gradually decreases. This is because the advantage of semisupervised learning becomes less pronounced when sufficient labeled data are available.

V. Conclusion

In this article, we present a Deform HGCN, which dynamically modulates vertex–hyperedge associations to enable long-range, high-order relational modeling under a controlled computational cost. By integrating Deform HGCN into a semisupervised teacher–student architecture for HSIC, our proposed DeformHG enables more effective utilization of unlabeled data. The experimental results across several benchmark datasets confirm the advantages of jointly integrating deformable mechanisms, high-order relational modeling, and robust semisupervised optimization strategies. However, deformable graph convolutional classification for HSI is still in its early stages. In the future, the following two directions merit further investigation: 1) optimizing Deform HGCN into more lightweight architectures to enhance their practicality and scalability in real-world HSI applications and 2) designing models that further alleviate reliance on labeled data, particularly for scenarios with extremely limited annotations.

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